



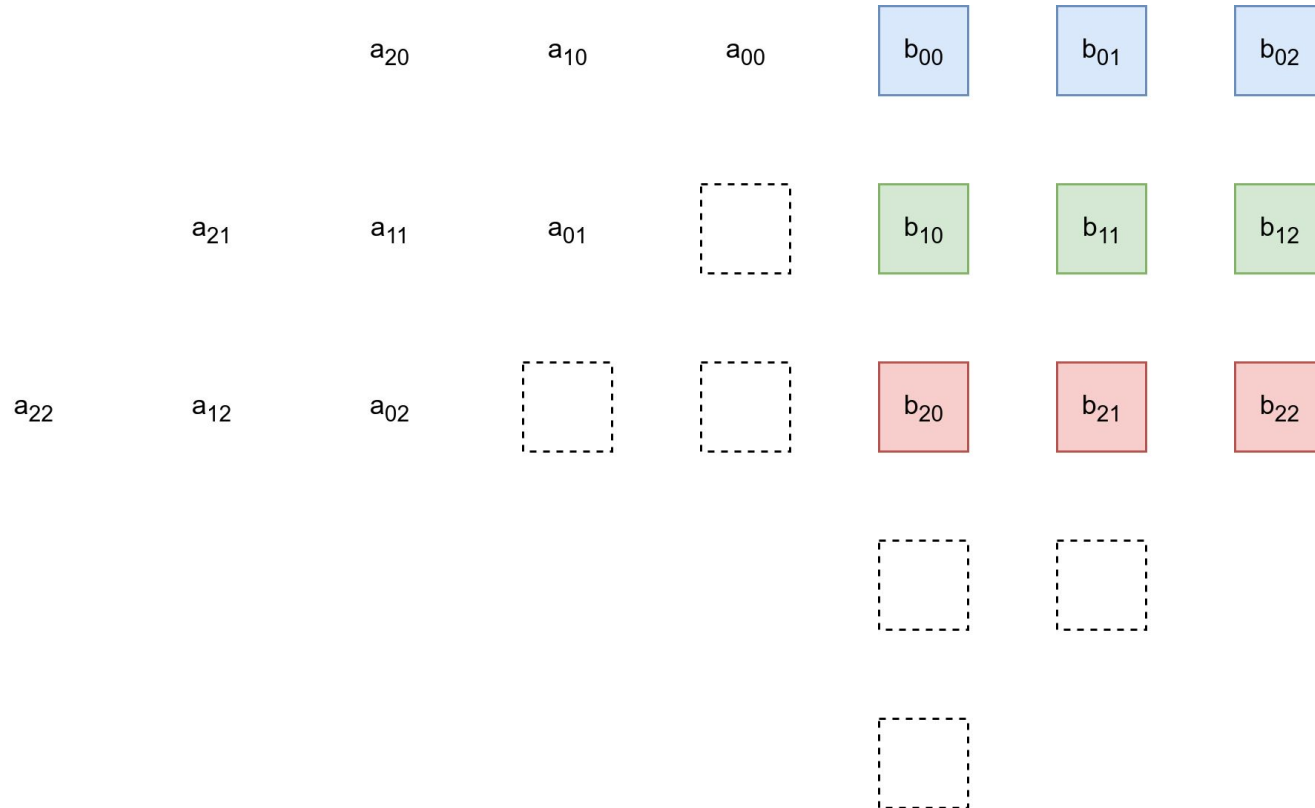
Систолический массив

Проектирование цифровых вычислительных систем

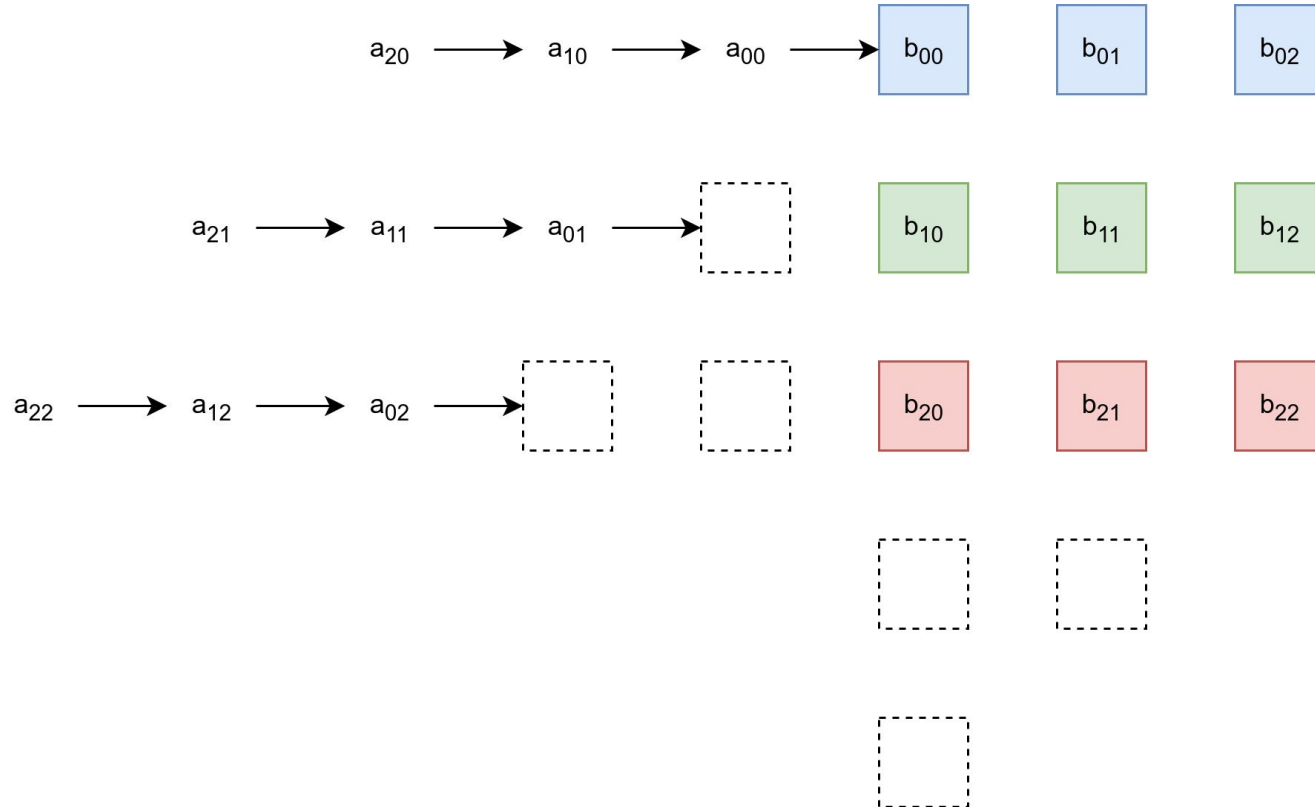
Прутьянов В. В.



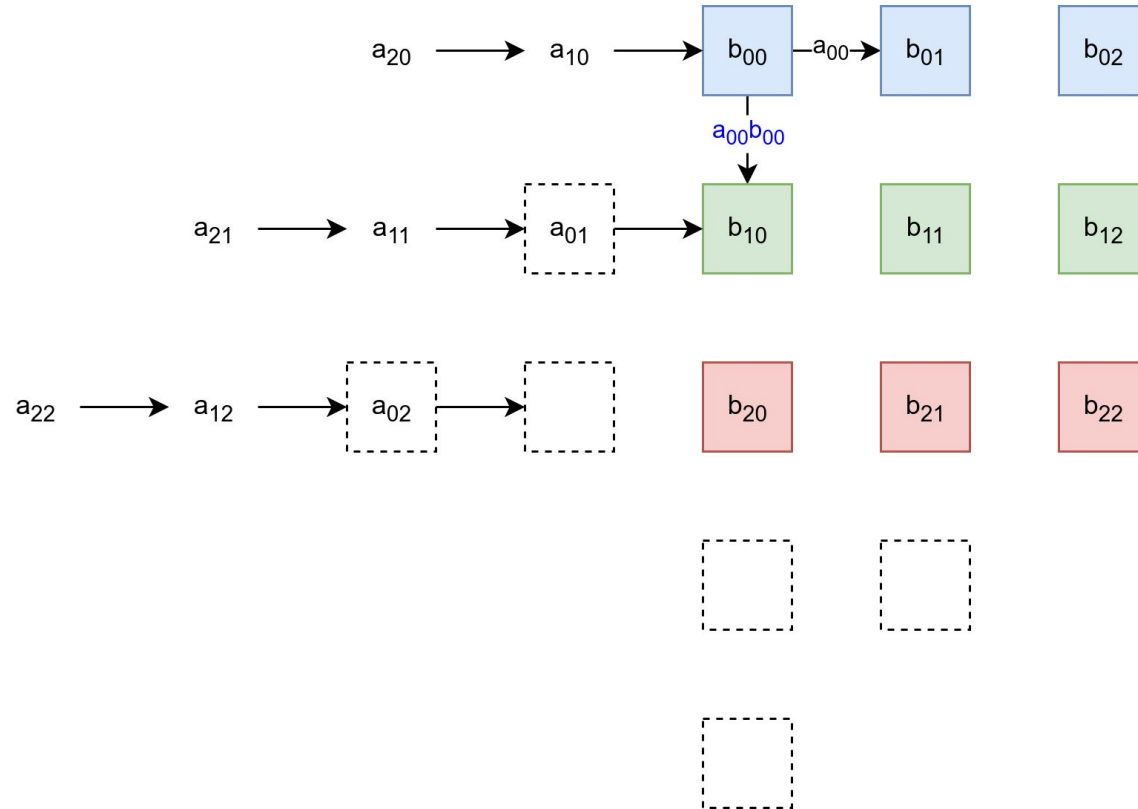
Systolic Array



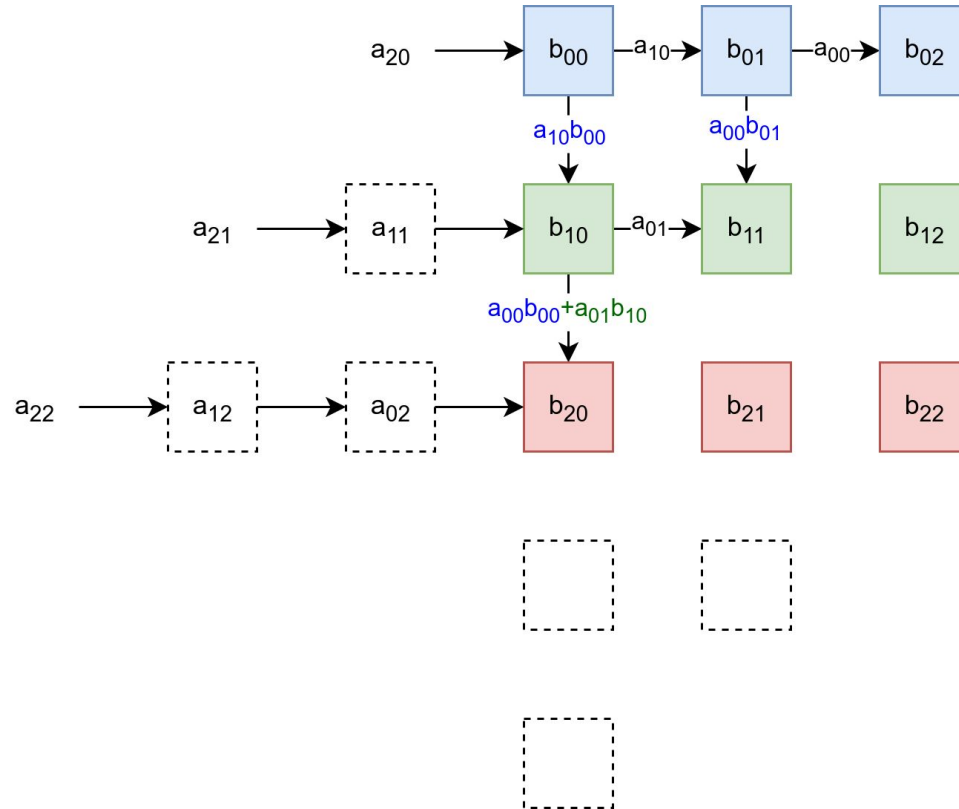
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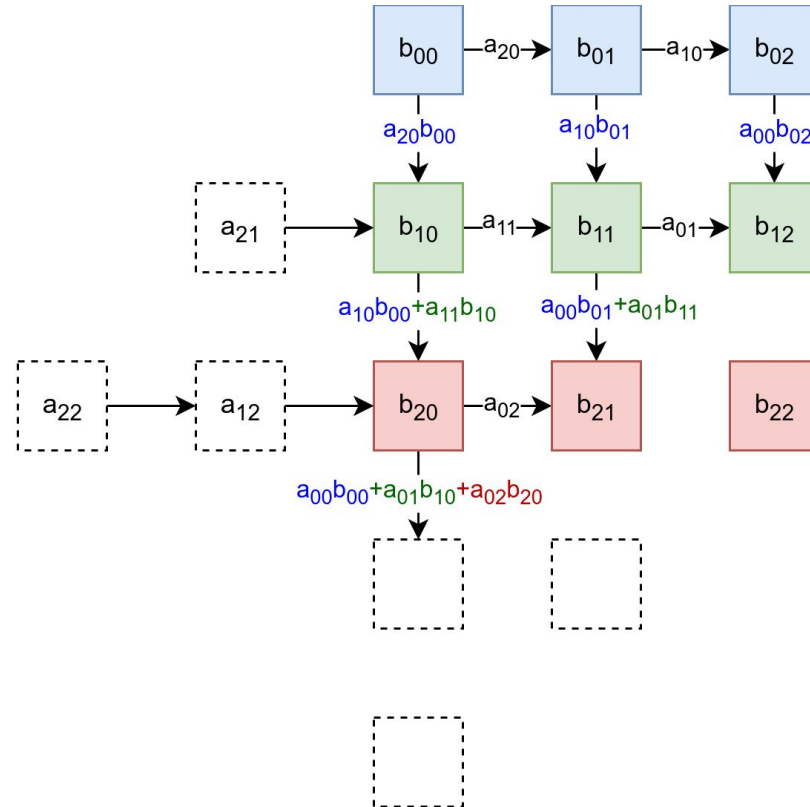
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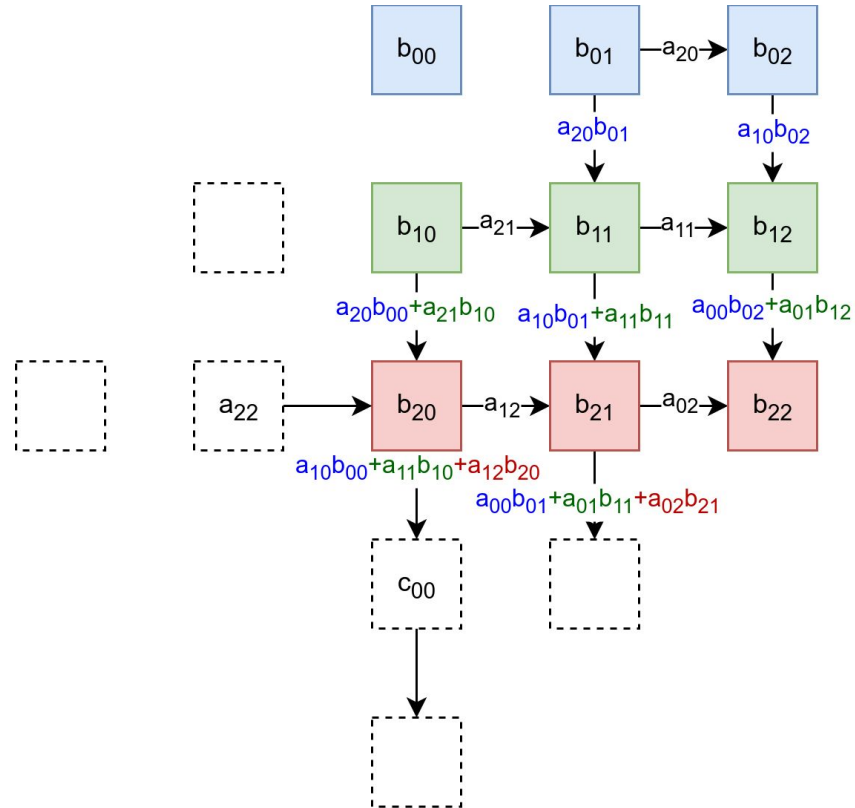
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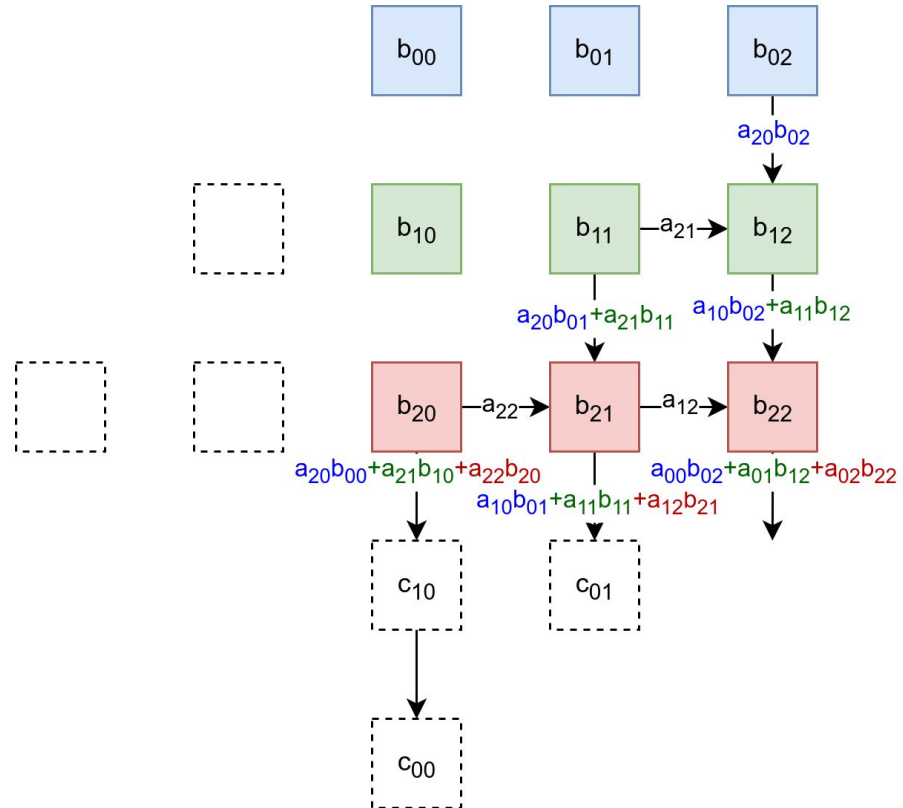
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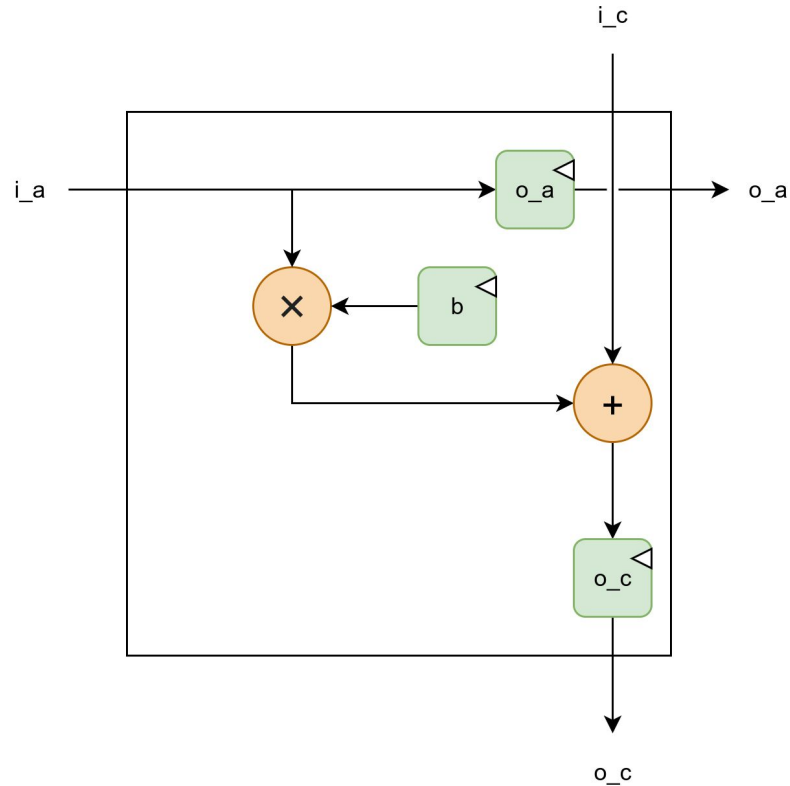
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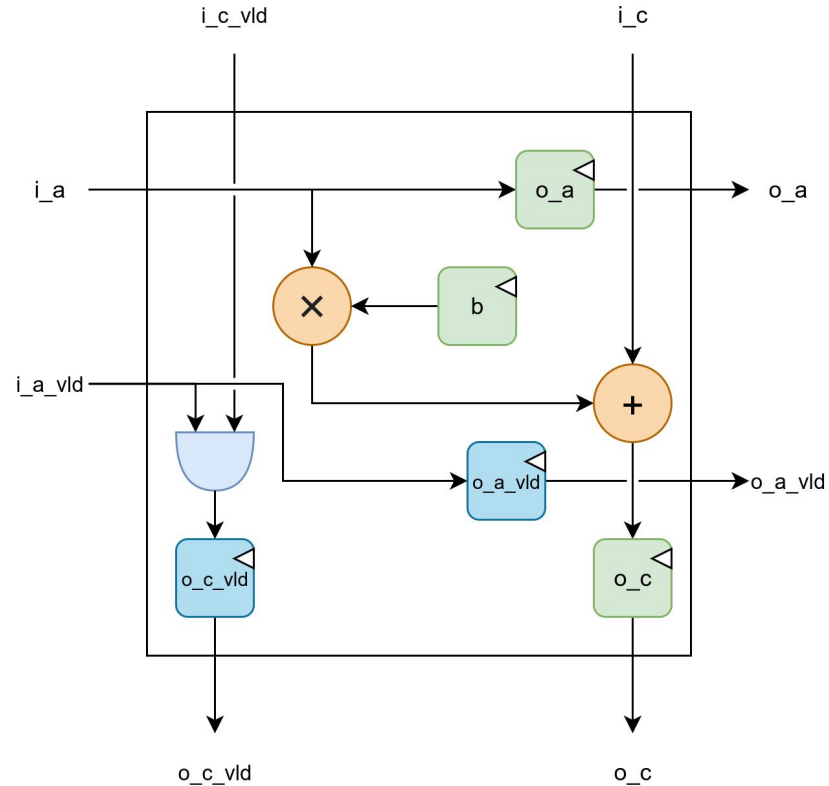
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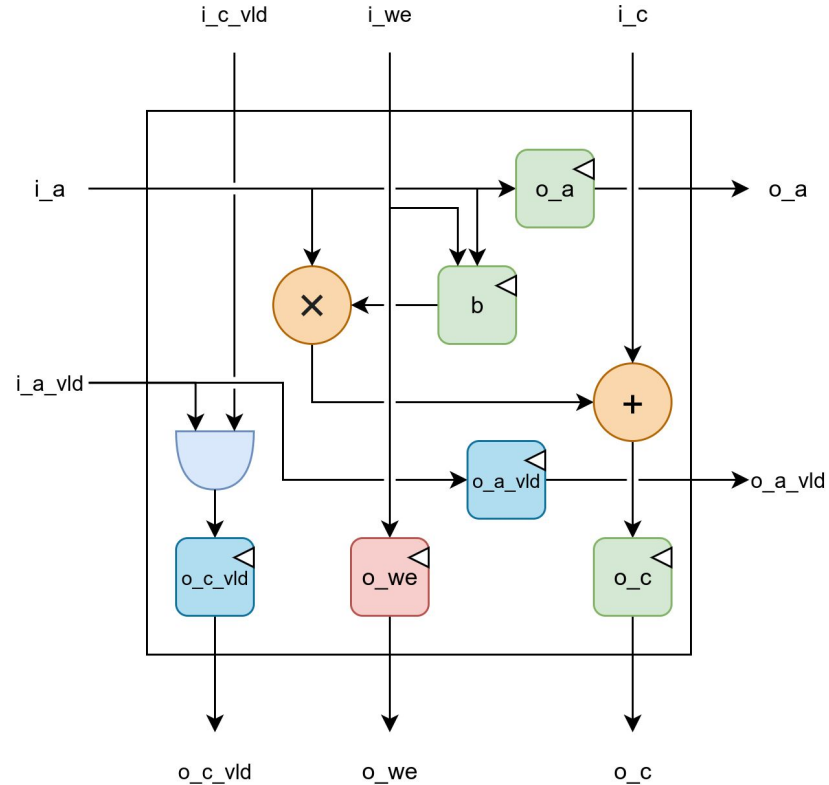
Processing Element: Data Flow



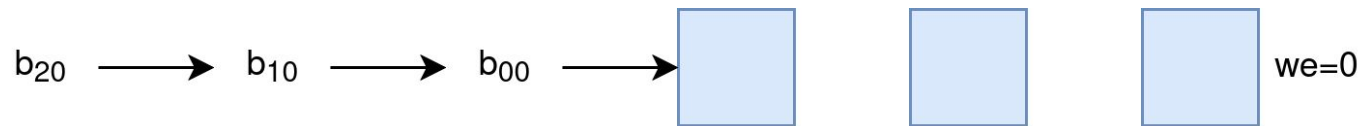
Processing Element: Control Flow



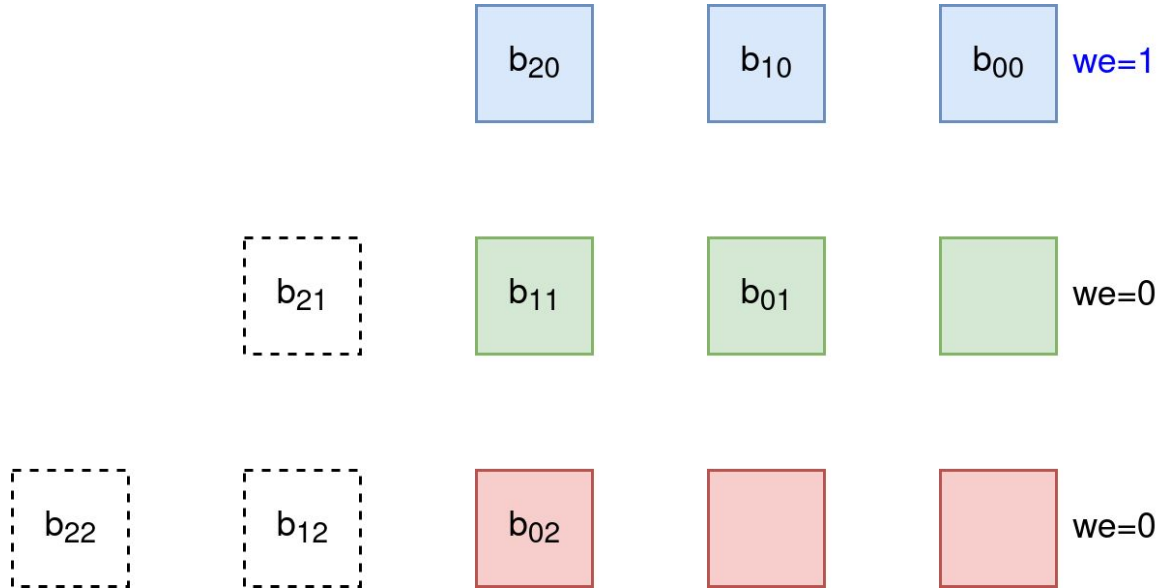
Processing Element: Write Enable



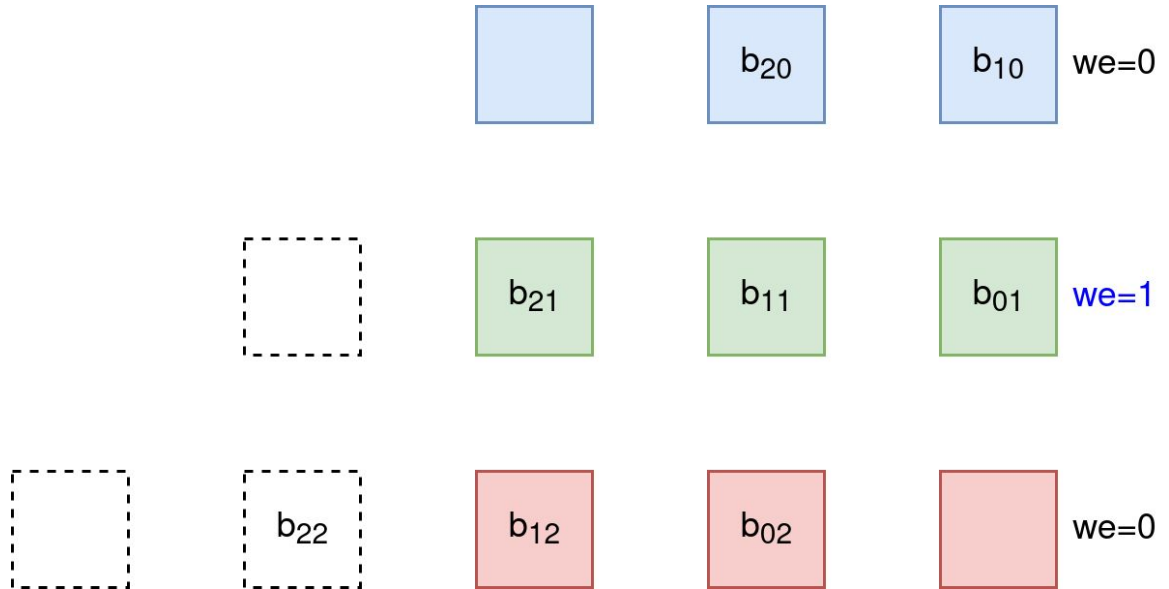
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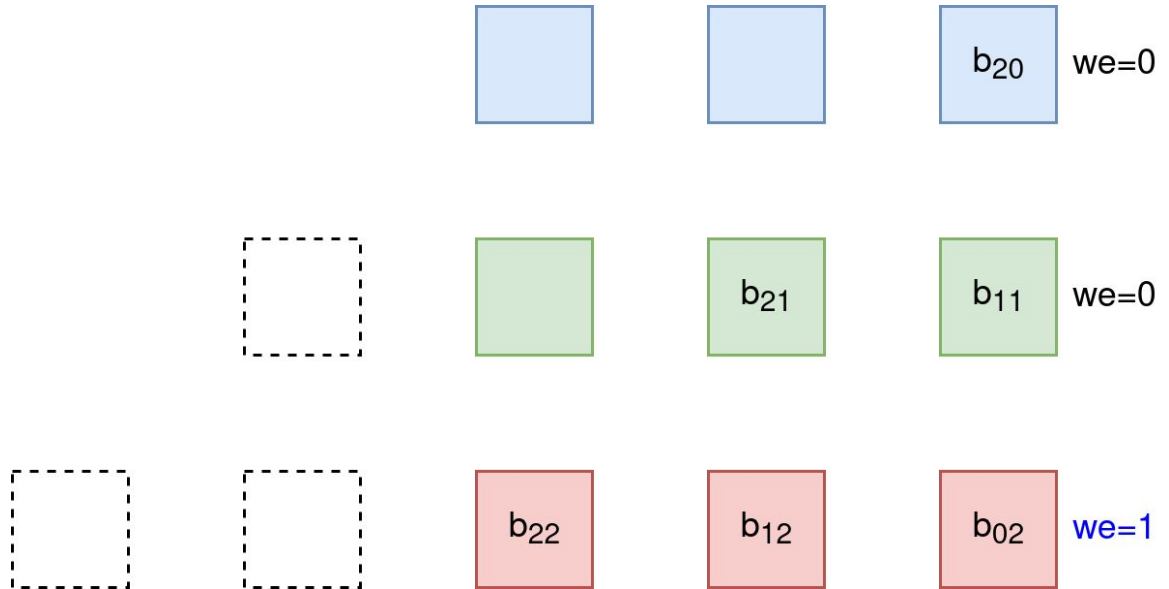
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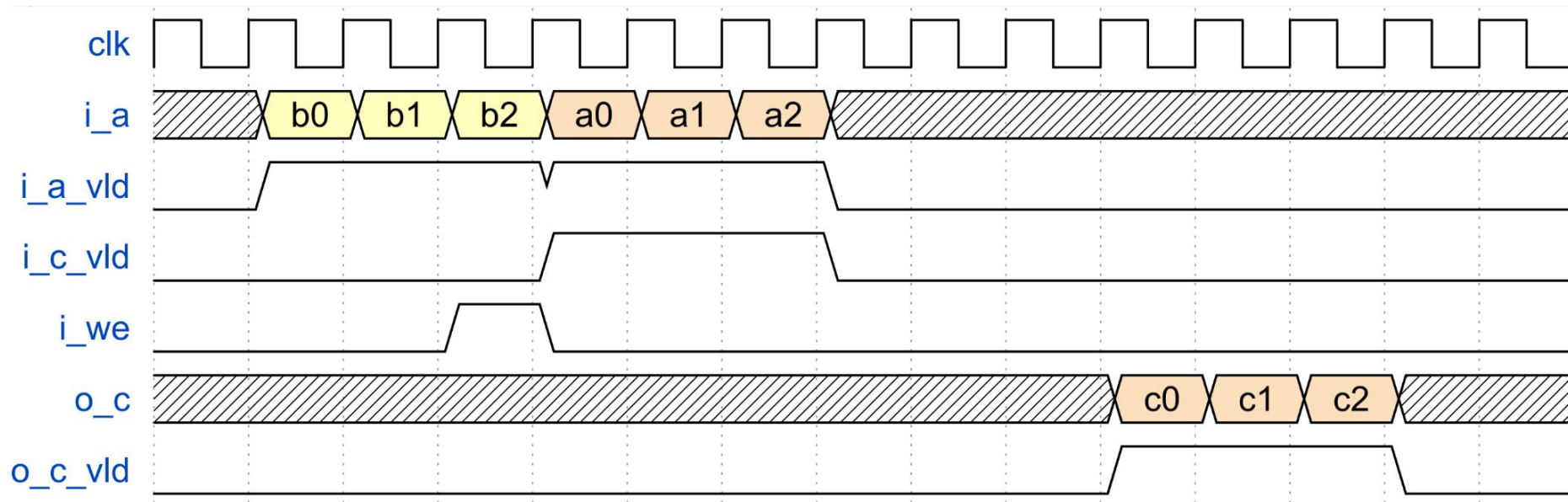
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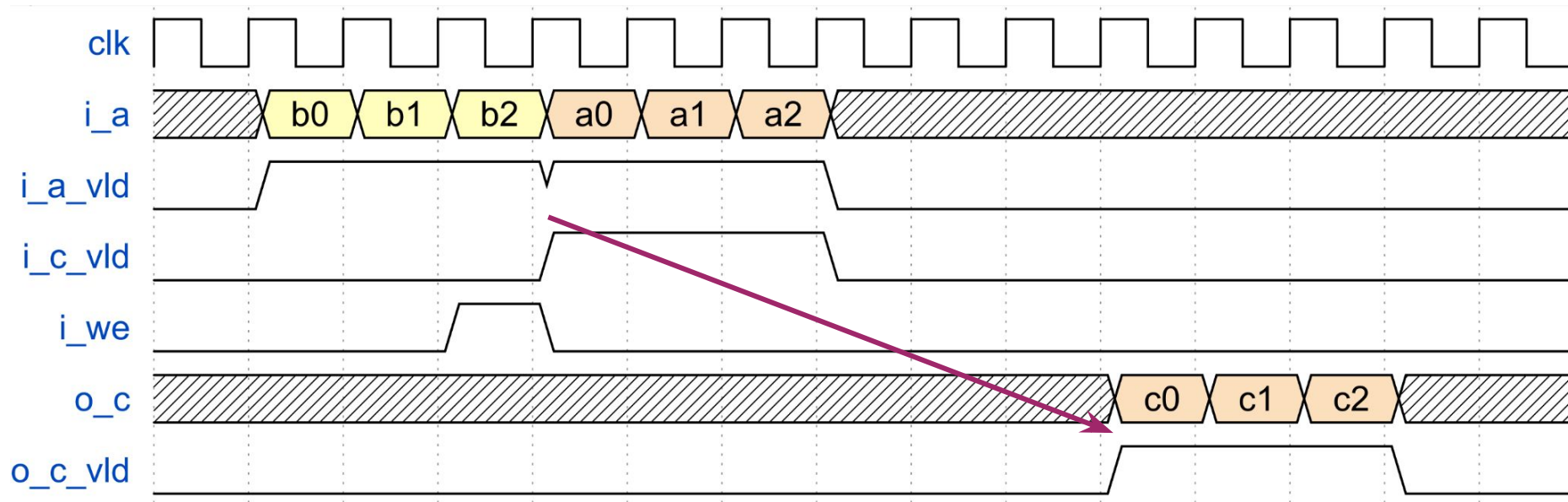
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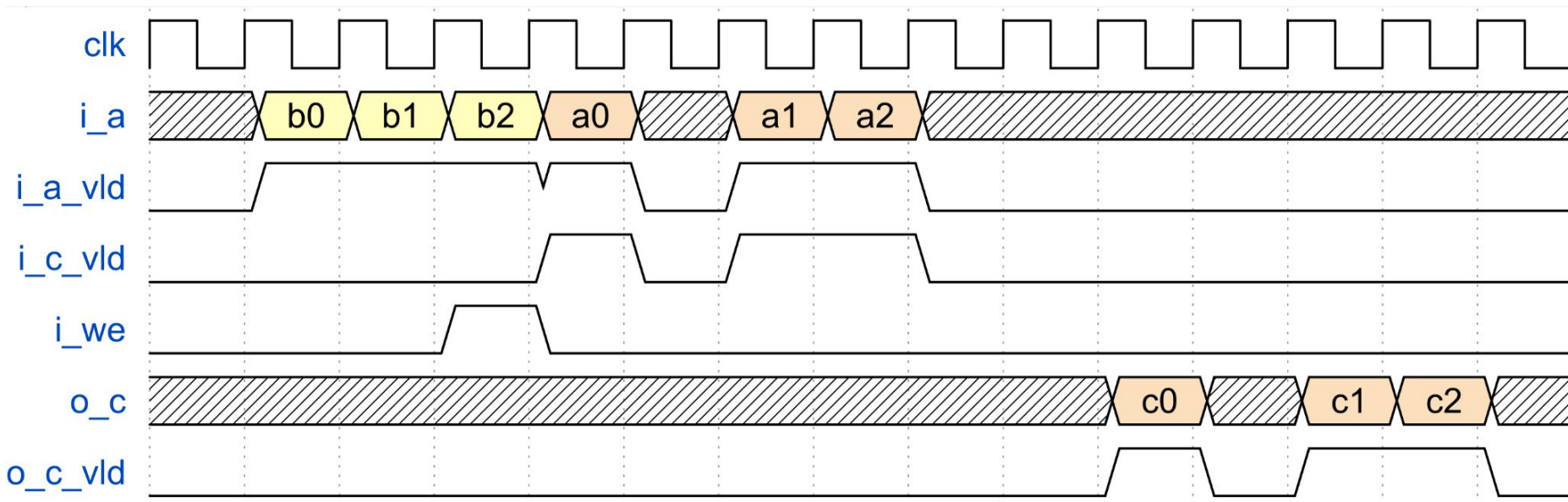
Systolic Array



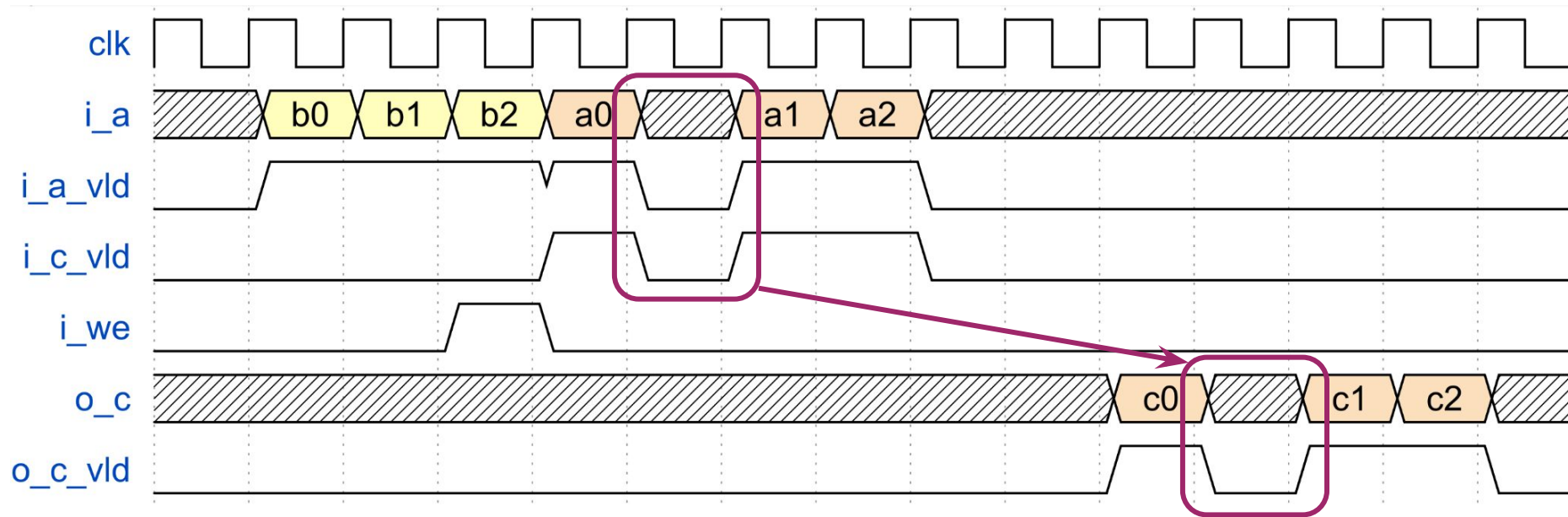
Systolic Array



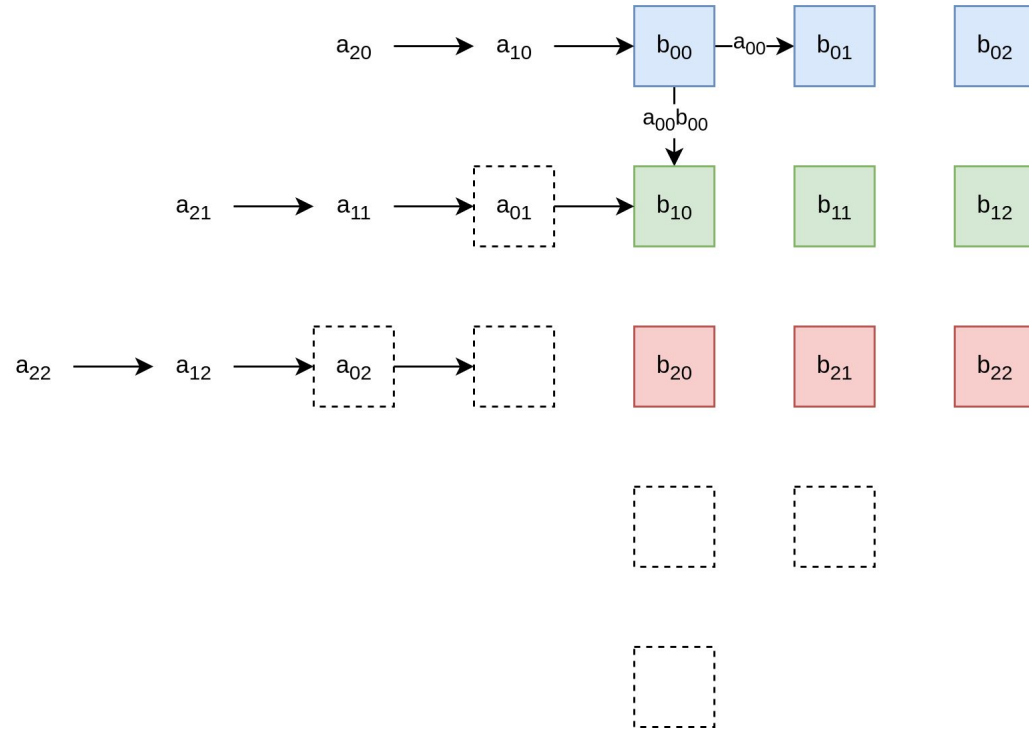
Systolic Array: Idle (A)



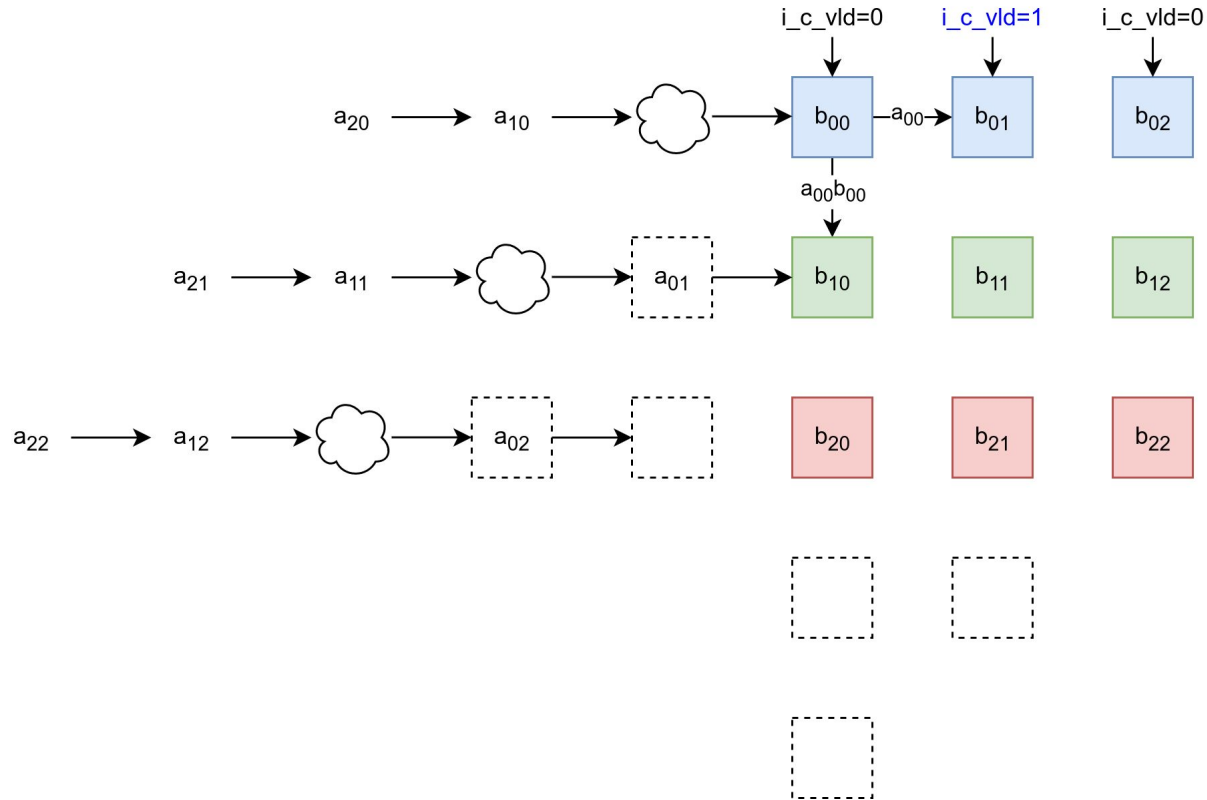
Systolic Array: Idle (A)



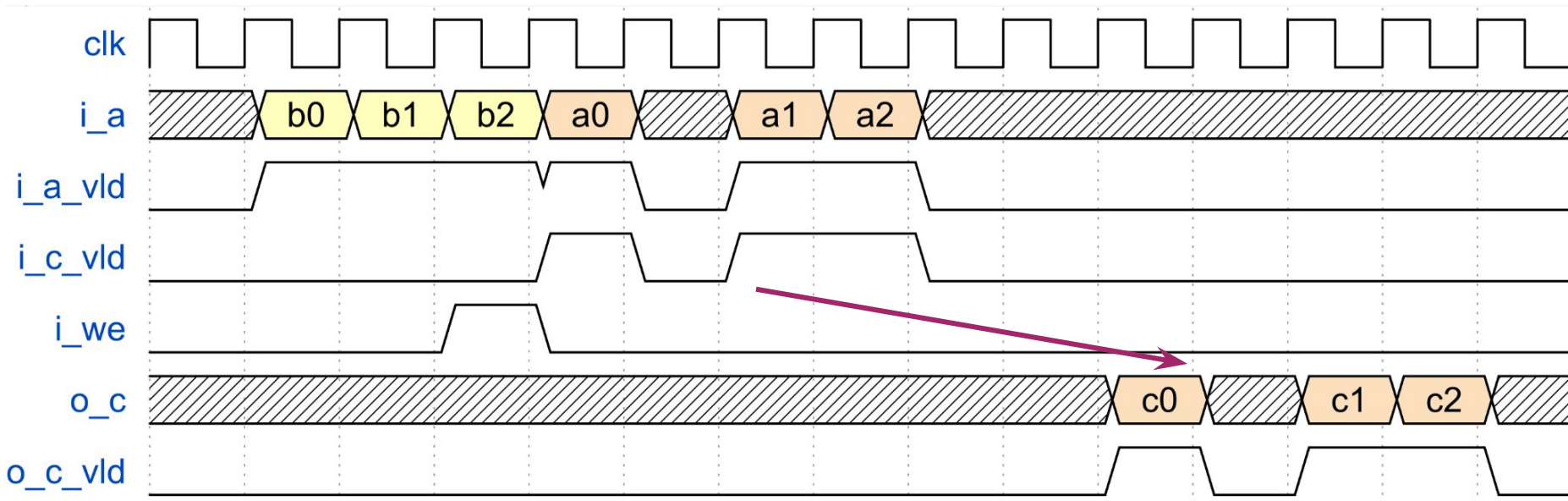
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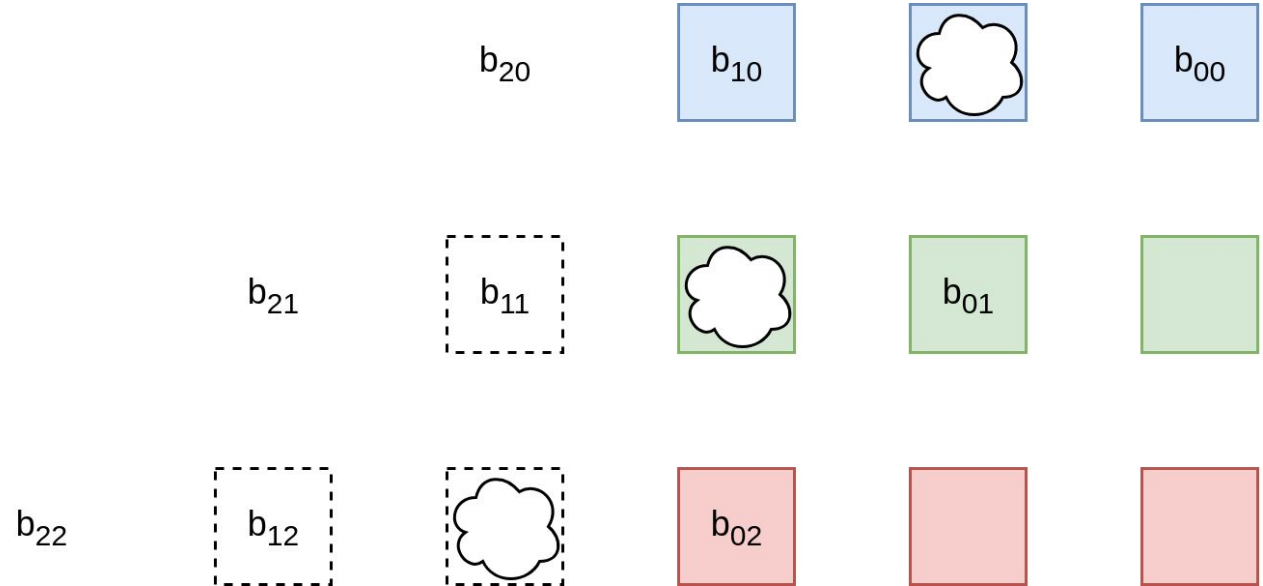
Systolic Array: Idle (A)



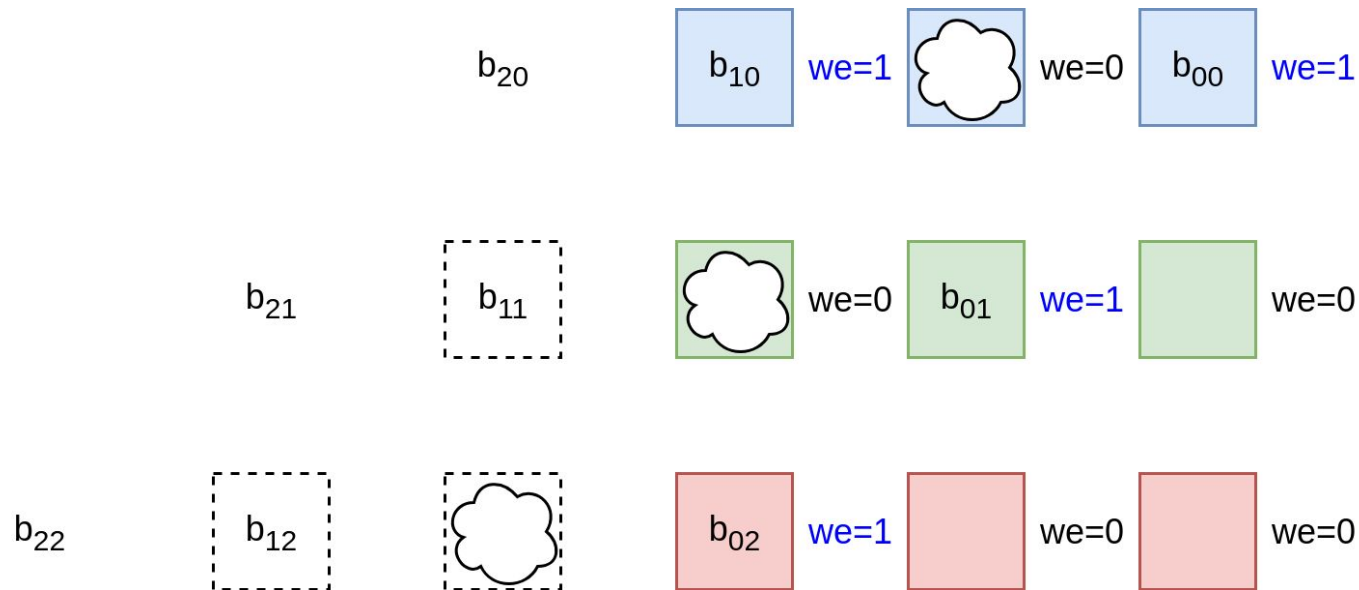
Systolic Array: Idle (A)



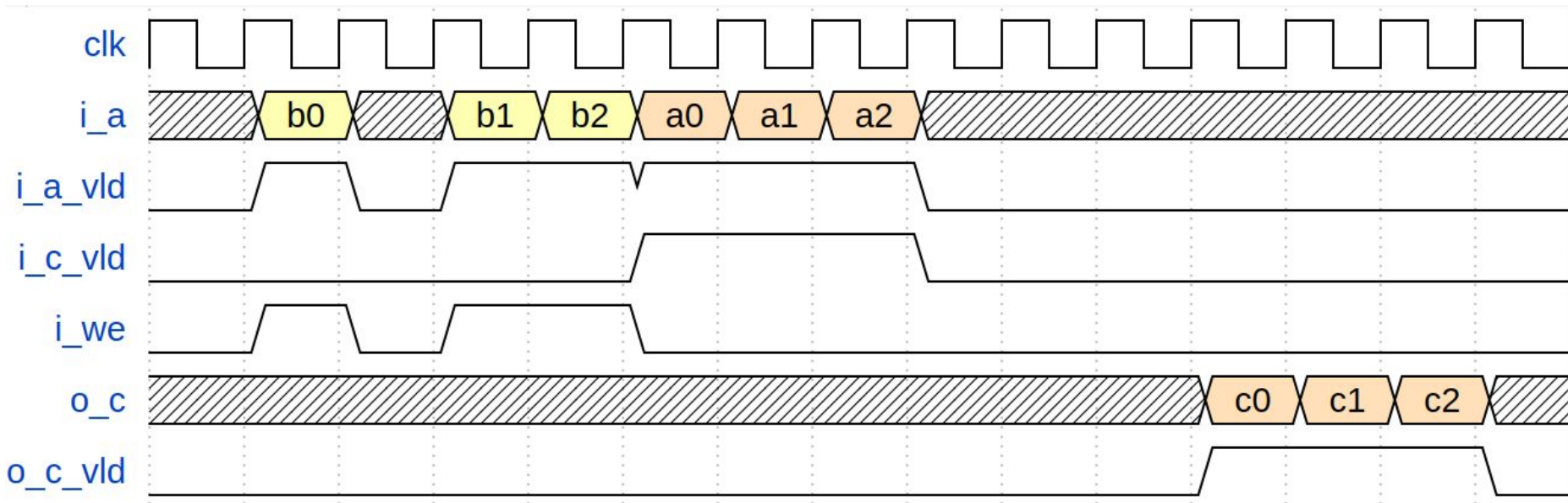
Systolic Array: Idle (B)



Systolic Array: Idle (B)



Systolic Array: Idle (B)



Задание

1. Processing Element, параметризованный разрядностью операндов ($WIDTH=16$)
2. Систолический массив, параметризованный размером ($SIZE=4$) и разрядностью
3. Сдвиговый регистр с valid-интерфейсами, параметризованный разрядностью
4. Систолический массив ($WIDTH=16, SIZE=4$) со сдвиговыми регистрами
5. DPI-тестбенч для систолического массива, проверяющий умножение матриц $m, n, k=4, 4, 4$
6. Систолический массив с поддержкой Idle по A и B матрицам

Литература

- Jouppi et al., “In-Datacenter Performance Analysis of a Tensor Processing Unit”, ISCA 2017
- H.T. Kung, “Why Systolic Architectures?,” IEEE Computer 1982